

Shared evidence • Constructed surface test

The friction surface lab

Name _____ Date _____

Same small wooden block and contact face; fixed low ramp and release mark; no push. Only the level test surface changes. Measure from ramp end to the same trailing block edge at stopping.

Surface	Run 1 (cm)	Run 2 (cm)	Run 3 (cm)
Card	58	60	62
Cloth	22	23	24
Rubber	14	15	16

Mean run-out: card 60 cm; cloth 23 cm; rubber 15 cm. These are summaries of supplied teaching data, not universal properties of every material pair. Friction opposes this block’s sliding.

W1. Which surface best fits a short model stopping zone? Use evidence.

W2. Why must we repair a run given an extra push?

Route 1 • Choose using the readings

Word bank: rubber / card / same / opposite / cm.

S1. The shortest run-out is on...

- A. Card
- B. Cloth
- C. Rubber

S2. Give one reading supporting your choice.

S3. The block slides right. Friction on it acts...

- A. Right
- B. Left

S4. Name two things the fair test keeps the same.

S5. Would you give one run an extra push? Why?

Route 2 • Write a surface recommendation

Recommend a material for this model stopping zone, using repeated evidence.

M1. State the changed factor and measured outcome.

M2. Compare card and rubber using their repeated ranges.

M3. Explain friction's direction and effect after release.

M4. Write a short method with a consistent release, distance rule and repeat.

Route 3 • Review the test quality

A designer claims: “Rubber always has the most friction, so our result proves the safest surface for every purpose.”

T1. Limit the claim to the actual object and setup.

T2. Explain why a different material pair or condition could need a new test.

T3. A model needs a long sliding distance instead. Which supplied surface fits that purpose?

T4. One cloth strip bunches up and stops the block. How should the team record and investigate this?

Exit and reflection

Point, draw, write or tell an adult.

E1. What does friction oppose in this test?

E2. Which supplied surface gives the shortest run-out?

E3. Name one start condition to keep the same.

R1. Which idea did you improve or defend? What evidence helped?

R2. What useful question would you investigate next?
